

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks:50

Annexure A

Coding challenge

Code of Conduct:

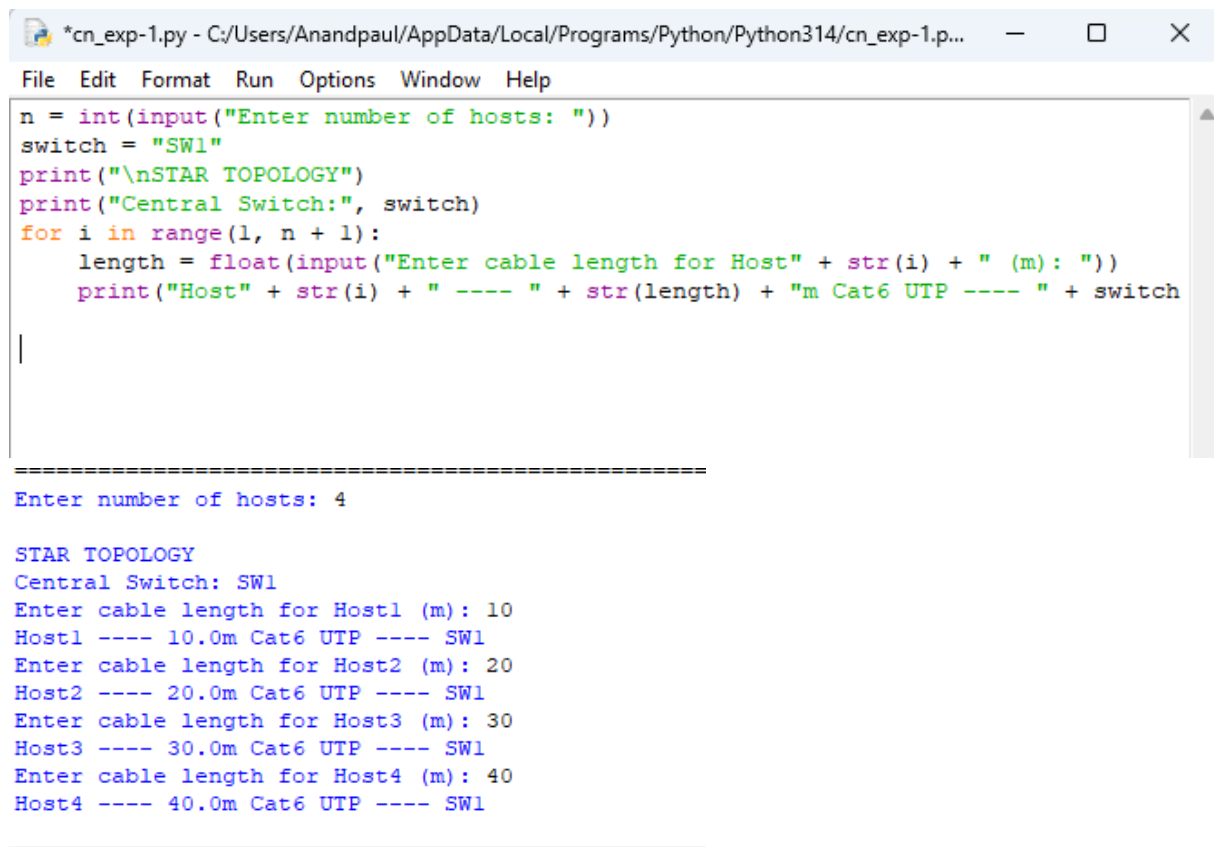
*I Anand Paul.D (192425207) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.
2. Code a function `validate_segment(length_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤ 100 m) and False with an error message otherwise. Include at least 3 test cases.
3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma_log.txt`.
4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.
5. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

1.Star Topology Generation using Python



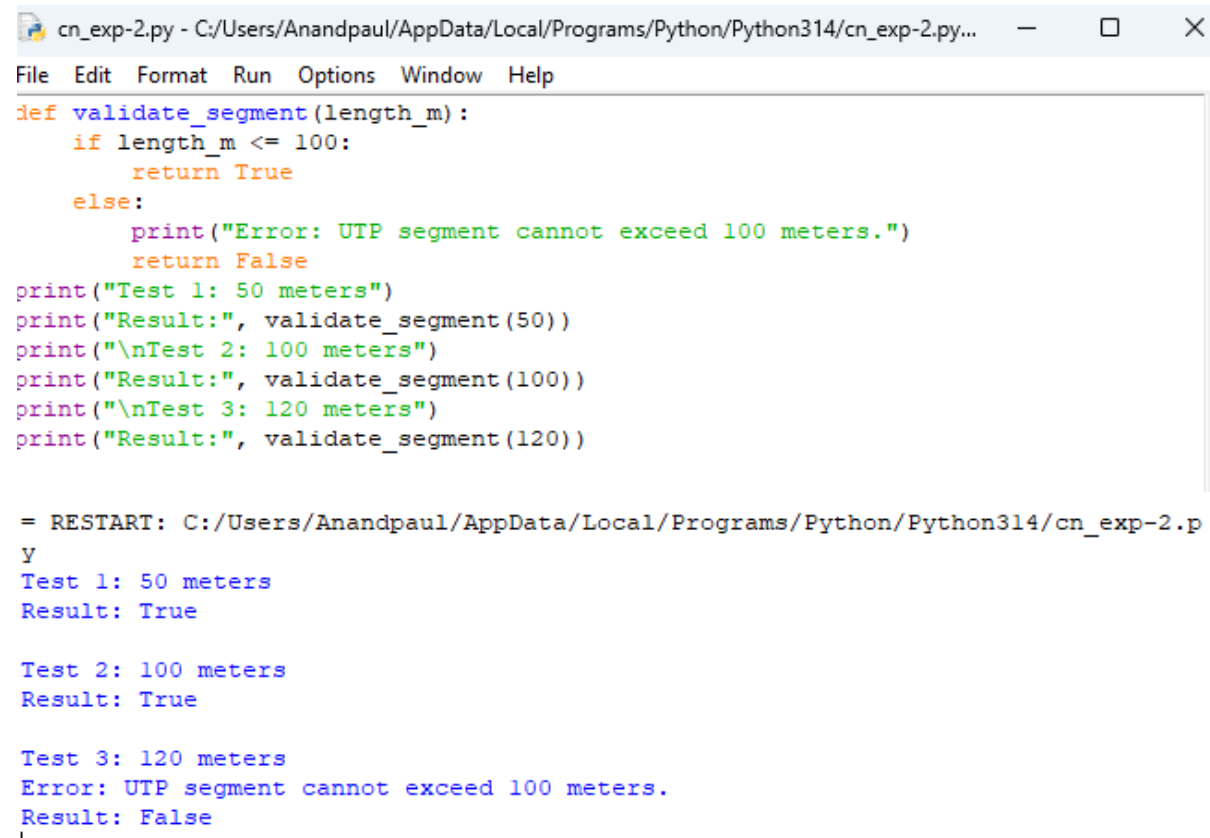
```
*cn_exp-1.py - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-1.p...
File Edit Format Run Options Window Help

n = int(input("Enter number of hosts: "))
switch = "SW1"
print("\nSTAR TOPOLOGY")
print("Central Switch:", switch)
for i in range(1, n + 1):
    length = float(input("Enter cable length for Host" + str(i) + " (m): "))
    print("Host" + str(i) + " ---- " + str(length) + "m Cat6 UTP ---- " + switch)

=====
Enter number of hosts: 4

STAR TOPOLOGY
Central Switch: SW1
Enter cable length for Host1 (m): 10
Host1 ---- 10.0m Cat6 UTP ---- SW1
Enter cable length for Host2 (m): 20
Host2 ---- 20.0m Cat6 UTP ---- SW1
Enter cable length for Host3 (m): 30
Host3 ---- 30.0m Cat6 UTP ---- SW1
Enter cable length for Host4 (m): 40
Host4 ---- 40.0m Cat6 UTP ---- SW1
```

2.UTP Cable Segment Length Validation



```
cn_exp-2.py - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-2.py...
File Edit Format Run Options Window Help

def validate_segment(length_m):
    if length_m <= 100:
        return True
    else:
        print("Error: UTP segment cannot exceed 100 meters.")
        return False
print("Test 1: 50 meters")
print("Result:", validate_segment(50))
print("\nTest 2: 100 meters")
print("Result:", validate_segment(100))
print("\nTest 3: 120 meters")
print("Result:", validate_segment(120))

= RESTART: C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-2.p
y
Test 1: 50 meters
Result: True

Test 2: 100 meters
Result: True

Test 3: 120 meters
Error: UTP segment cannot exceed 100 meters.
Result: False
|
```

3.CSMA/CD Simulation on Star Topolog

```
cn_exp-3.py - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-3.py...
File Edit Format Run Options Window Help

import random
hosts = ["Host1", "Host2", "Host3", "Host4"]
log = open("csma_log.txt", "w")
print("CSMA/CD SIMULATION")
print("-----")
for i in range(1, 6):
    host = random.choice(hosts)
    print("\nRound", i)
    print(host, "is trying to transmit")
    log.write("Round " + str(i) + "\n")
    log.write(host + " is trying to transmit\n")
    collision = random.choice([True, False])
    if collision:
        backoff = random.randint(1, 5)
        print("Collision detected!")
        print("Back-off interval:", backoff)
        log.write("Collision detected!\n")
        log.write("Back-off interval: " + str(backoff) + "\n")
    else:
        print("Transmission successful.")
        log.write("Transmission successful.\n")
log.close()
print("\nSimulation completed.")
print("Log saved as csma_log.txt")

=====
CSMA/CD SIMULATION
-----

Round 1
Host1 is trying to transmit
Transmission successful.

Round 2
Host4 is trying to transmit
Collision detected!
Back-off interval: 2

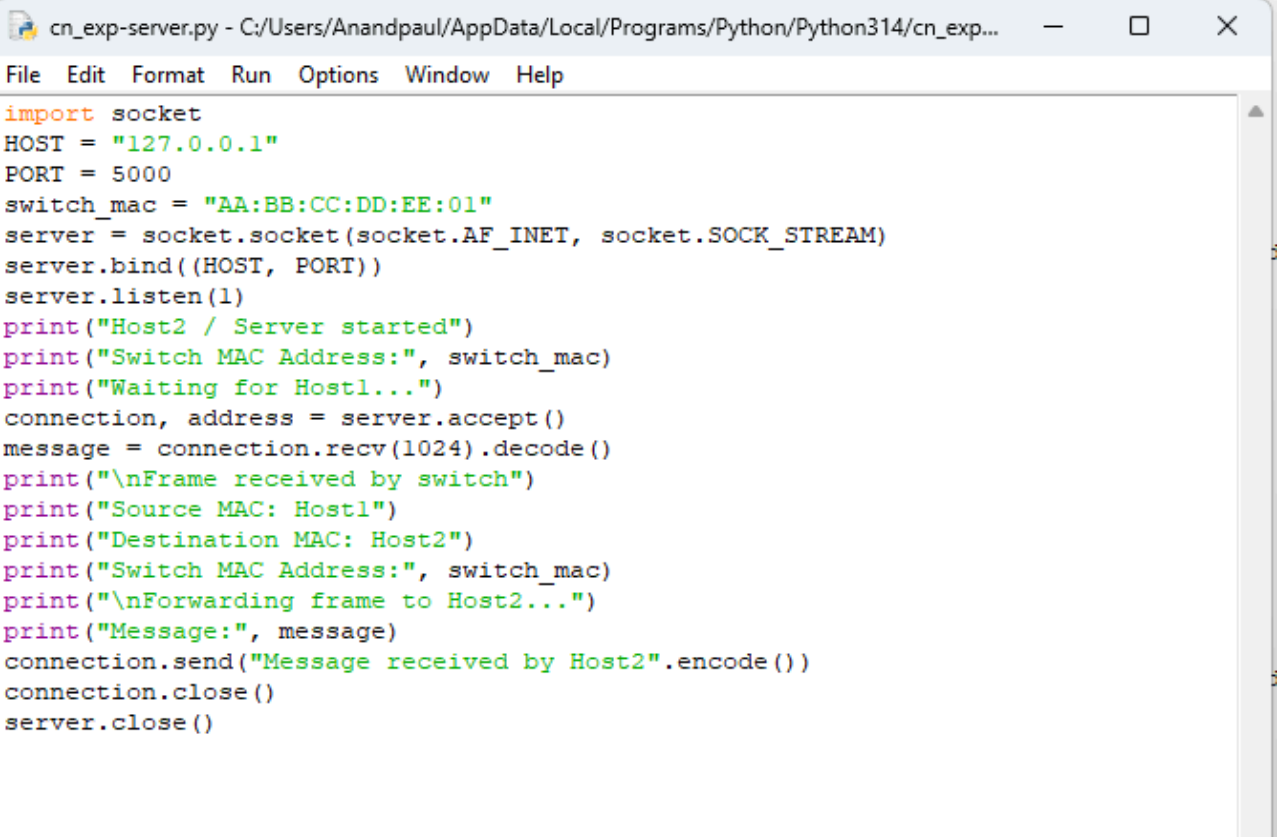
Round 3
Host4 is trying to transmit
Transmission successful.

Round 4
Host4 is trying to transmit
Transmission successful.

Round 5
Host2 is trying to transmit
Transmission successful.

Simulation completed.
Log saved as csma_log.txt
```

4.Client-Server Communication using Python Sockets



```
cn_exp-server.py - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp...
File Edit Format Run Options Window Help

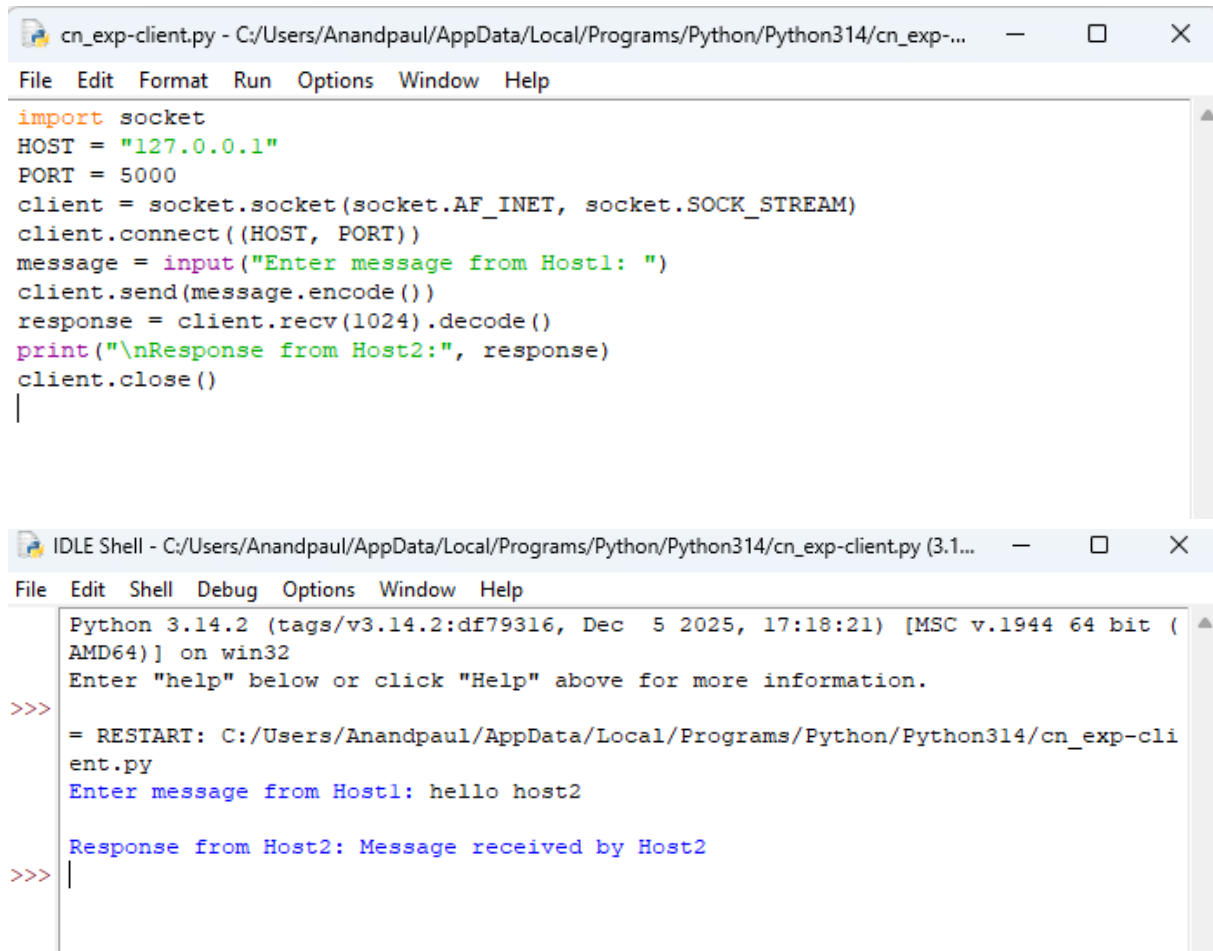
import socket
HOST = "127.0.0.1"
PORT = 5000
switch_mac = "AA:BB:CC:DD:EE:01"
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((HOST, PORT))
server.listen(1)
print("Host2 / Server started")
print("Switch MAC Address:", switch_mac)
print("Waiting for Host1...")
connection, address = server.accept()
message = connection.recv(1024).decode()
print("\nFrame received by switch")
print("Source MAC: Host1")
print("Destination MAC: Host2")
print("Switch MAC Address:", switch_mac)
print("\nForwarding frame to Host2...")
print("Message:", message)
connection.send("Message received by Host2".encode())
connection.close()
server.close()
```

```
Host2 / Server started
Switch MAC Address: AA:BB:CC:DD:EE:01
Waiting for Host1...

Frame received by switch
Source MAC: Host1
Destination MAC: Host2
Switch MAC Address: AA:BB:CC:DD:EE:01

Forwarding frame to Host2...
Message: hello host2
```

4.1 Client



The image shows two windows from a Python IDE. The top window, titled 'cn_exp-client.py', contains the following Python code:

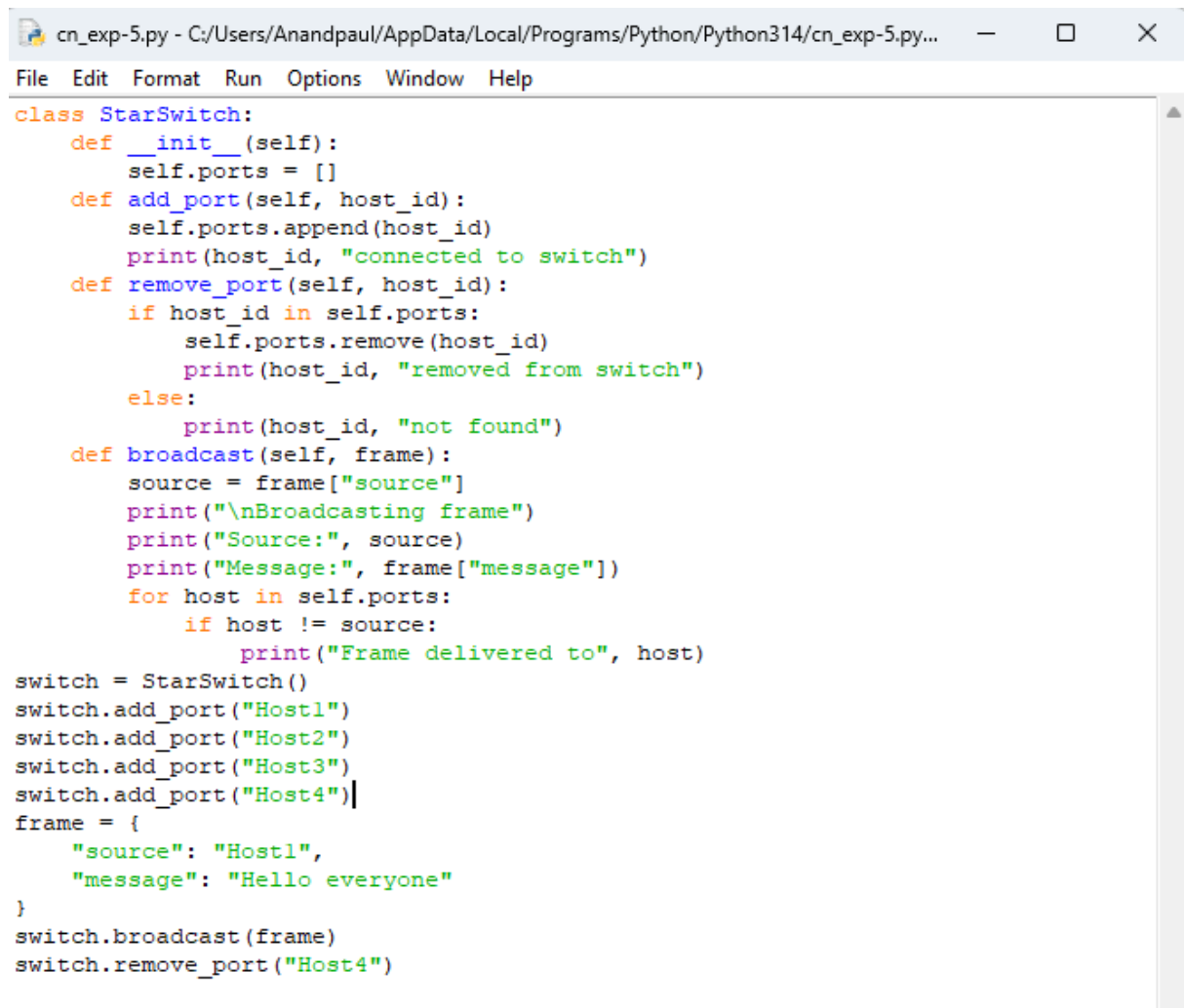
```
import socket
HOST = "127.0.0.1"
PORT = 5000
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((HOST, PORT))
message = input("Enter message from Host1: ")
client.send(message.encode())
response = client.recv(1024).decode()
print("\nResponse from Host2:", response)
client.close()
```

The bottom window, titled 'IDLE Shell - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-client.py (3.1...', shows the execution output:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-client.py
Enter message from Host1: hello host2

Response from Host2: Message received by Host2
>>>
```

5.Star Switch Simulation using Python Class



```
cn_exp-5.py - C:/Users/Anandpaul/AppData/Local/Programs/Python/Python314/cn_exp-5.py...
File Edit Format Run Options Window Help

class StarSwitch:
    def __init__(self):
        self.ports = []
    def add_port(self, host_id):
        self.ports.append(host_id)
        print(host_id, "connected to switch")
    def remove_port(self, host_id):
        if host_id in self.ports:
            self.ports.remove(host_id)
            print(host_id, "removed from switch")
        else:
            print(host_id, "not found")
    def broadcast(self, frame):
        source = frame["source"]
        print("\nBroadcasting frame")
        print("Source:", source)
        print("Message:", frame["message"])
        for host in self.ports:
            if host != source:
                print("Frame delivered to", host)

switch = StarSwitch()
switch.add_port("Host1")
switch.add_port("Host2")
switch.add_port("Host3")
switch.add_port("Host4")
frame = {
    "source": "Host1",
    "message": "Hello everyone"
}
switch.broadcast(frame)
switch.remove_port("Host4")
```

```
Host1 connected to switch
Host2 connected to switch
Host3 connected to switch
Host4 connected to switch
```

```
Broadcasting frame
Source: Host1
Message: Hello everyone
Frame delivered to Host2
Frame delivered to Host3
Frame delivered to Host4
Host4 removed from switch
```